Galaba Vamsi

Software Engineer | AI/ML | Computer Vision | Systems
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PROFESSIONAL SUMMARY

Mechatronics undergraduate at IIT Bhilai focused on software engineering, AI/ML, computer vision, robotics, and GPU systems. Builds full-stack products, open-source tooling, simulators, and visual applications with Python, TypeScript, React, PyTorch, CUDA, and cloud platforms.

EXPERIENCE

Open-Source Contributor - Inspect Robots 2026
Robocurve (YC Summer 2026) | Repository | inspect-robots-wandb on PyPI
- Shipped inspect-robots-wandb v0.1.0, a Weights & Biases logging sink (100% coverage, CI on Python 3.10-3.13, PyPI trusted publishing).
- 2 merged PRs: #476 configurable LLM retry policy over 5 wire protocols; #475 unknown git state no longer logged as clean.
- 3 open PRs awaiting merge: #514 crash-safe checkpoint/resume for robot eval sets; #550 `on_eval_error` sink hook; #551 plugin docs.

Open-Source Contributor - Cerulion 2026
Rust robotics middleware | Repository
- 5 merged PRs; none open: #216 safe `ros2 attach --dry-run` outside a workspace; #226 fail-closed `CERULION_NETWORK` docs; #220 workspace-root comments; #222 ASCII punctuation; #224 removed stale `USER_API` references.

Open-Source Contributor - HFlow 2026
Hebbian Robotics (YC S26) | HFlow repository
- 8 maintainer-reviewed PRs merged. Latest: #657 types Arrow columns from all messages, preserving late fields and rejecting incompatible shapes; #598 refused multi-channel topics before episode creation; #400 refused LeRobot depth videos to preserve 12-bit depth.
- #367 benchmarked 3 cold 1080p30 checks (900 frames each; 4.897 s median); #350 guarded paginated LeRobot traversal; #353 fixed cache collisions; #360 centralized URI parsing; #362 matched EgoSuite labels by source provenance.

Research Intern - 6G RIS Simulator May 2025 - Aug 2025
IIT Bhilai | Advisor: Dr. Arzad Alam
- Built a zero-budget open-source Python RIS simulator with USRP-style interfaces for LoS/NLoS environments (RIM-SIM v2: 6 scenarios, 20 Hz WebSocket feed, 103 tests). [GitHub]
- Designed a 2-layer RIS PCB workflow for RF energy harvesting; co-authored 2 papers (INDICON 2025, Ad Hoc Networks 2026); first-author manuscript in preparation.

Marketing Intern - Campus CEO Project May 2024 - Jul 2024
Swiggy Ltd
- Managed a campaign with 15 social media posts; contributed to campus marketing initiatives and execution.

PROJECTS

Human Slop - Co-Founder 2026 - Present
Built and deployed an anti-AI web and Android platform with React, PostgreSQL, and AWS; implemented hardware-bound biometric authentication and real-time typing forensics. [Website] | [App]

Charter - Agentic Commerce
Agentic-commerce platform (TypeScript, Fastify, Postgres, React) with chat, voice, and MCP interfaces, 10 MCP tools over a shared core, frozen INR quotes, and RLS. [GitHub]

Pneumatic Sorting Digital Twin
Built an open-source Unity digital twin for modular electropneumatic automation with offline sorting and stamping kits, HMI flows, telemetry, and replay. [GitHub]

Quarter-Car Active Suspension - Paper Reproduction 2026
Reproduced an IEEE TSMC 2022 approximation-free suspension controller in MATLAB within 7% of the paper's Case-10 result; stress-tested it with stability analysis, LQR/skyhook baselines, a hold-out test, and an anti-windup extension. [Site] | [GitHub]

Whiteboard Cleaning Robot - ROS 2 2025
3-DOF Cartesian robot in ROS 2 Jazzy and Gazebo Harmonic: custom URDF, ros2_control, raster coverage path with 10 cm overlap, and logged coverage analysis. [GitHub]

Arista Networks Wi-Fi Optimization System - Inter IIT 5th Rank 2025
Client-aware Wi-Fi RRM in NS-3: 3 RF metrics, 100,000 I/Q samples into 16 classes (STFT, DBSCAN, Dual-CNN), 5 APs and 50 clients with IEEE 802.11k/v. [Slides]

Electron-GNN - ML-Accelerated Quantum Spectroscopy 2026
V3 two-tower GATv2 predicting molecular absorption spectra from geometry; 2 molecules, 39 and 55 target peaks. [GitHub]. Faculty advisor: Dr. Soumajit Pramanik.

Open-PyFX - GPU-Accelerated Video Effects 2025
Python and GLSL GPU video effects suite: CRT emulation, ASCII edge detection, 3D LUTs, 3840x2160 export. [GitHub]

Lunar Crater Detection - DEM Terrain Analysis 2025
Crater extraction from DEMs: F1 0.839, about 4x preprocessing speedup with CuPy/CUDA, 77 tests. [GitHub]

AntennaNet - Inverse Electromagnetic Design 2025
KD-tree spectral anchoring for inverse EM design over 144D search spaces (Python, MEEP, OpenMPI); about 2,000 indexed spectra. [GitHub]

Amazon ML Challenge - Top 3.5% 2025
Multimodal price prediction with Qwen 2-VL 7B and a ViT/RoBERTa two-tower model; 75,000 data pairs; SMAPE 49.0.

SKILLS

Languages: Python, Rust, C/C++, TypeScript, JavaScript, SQL, GLSL, CUDA.
AI/ML: PyTorch, Graph Neural Networks (GATv2), Safe Reinforcement Learning, Transformers, OpenCV, CuPy, NumPy.
Software and Systems: React, React Native, Flutter, FastAPI, Node.js, REST APIs, WebSockets, PostgreSQL, AWS, Supabase, Docker, ROS 2, Gazebo, MuJoCo, MATLAB, MEEP, OpenMPI, SDR/USRP.
Developer Tools: Git, Linux, GitHub Actions, CI/CD, pytest, Ruff, mypy, clippy, PyPI publishing, static analysis, regression testing.

EDUCATION

B.Tech. in Mechatronics Aug 2023 - May 2027
Indian Institute of Technology Bhilai
Coursework: Data Structures and Algorithms, Object-Oriented Programming, Deep Learning for Computer Vision, AI and Machine Learning, Digital Image Processing, Optimal Control.

PUBLICATIONS

Co-author: *Experience with RF Energy Harvesting-Driven Self-Powered RIS*, IEEE INDICON 2025. [Link]
Co-author: *A systematic literature review on simulation models and deployments for reconfigurable intelligent surfaces*, Ad Hoc Networks (Elsevier), 2026. [Link]
First author (manuscript, arXiv preprint in preparation): *An Open Emulator for Smart Radio Environments*. [Draft]
Co-author (submitted, under review): *Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI*, CoRL 2026 Workshop SPAIS. [Framework]

ACHIEVEMENTS

1st Place: IIT Bhilai Web Development Competition.
5th Rank: Arista Networks Wi-Fi Challenge, Inter IIT Tech Meet 14.0.
AIR 200: Amazon ML Challenge 2026 (top 3.5% in 2025).
YC Startup School: Human Slop accepted.
Top 10 Finalist: Toyota Hackathon.